Axial Flux Electrical Machine Design Report

Automated Design Summary Generated via Python Simulation Suite

Simulation Solver:	Three dimension mesh based generated reluctance network (MBGRN)
Author / Lead Engineer:	Dat Van Vu

Project Description:

This simulation report details the electromagnetic performance, validation parameters, and core geometric data of an axial flux electrical machine design. All transient waveforms and performance metrics are processed and validated through the automated simulation pipeline.

1. General Specifications & Materials

Parameter Description	Value / Assignment
Machine Topology Classification	axial_flux_motor_type_1
Operational Shaft Rotor Speed	3000 RPM
Stator Winding Excitation (I RMS)	10.0 A
Current Advance Angle	0.0°
Core Stator Iron Material	steel_1008
Rotor Permanent Magnet Grade	NdFe30

2. Machine Core Geometry Parameters

Geometric Attribute (Stator)	Value	Geometric Attribute (Rotor)	Value
Stator Armature Outer Diameter	150.0 mm	Rotor Outer Diameter	150.0 mm
Stator Bore Internal Diameter	70.0 mm	Main Operational Airgap Length	1.50 mm
Total Slots Count	30	Total Pole Count	20
Armature Slot Width / Depth	5.0 / 15.0 mm	Permanent Magnet Arc	160° mechanical
Slot Opening Width	2.0 mm	Permanent Magnet Axial Length	3.0 mm
Active Axial Stator Core Length	25.0 mm	Rotor Yoke Structural Backing Length	6.0 mm

3. Armature Winding Configuration

Winding Specification	Configuration Setting
Number of Phases (m)	3
Total Series Turns Per Coil	20
Coil Throw Interval (Slots)	1
Winding Layer Assignment Count	2
Parallel Circuit Paths Count	1

4. Performance Metrics Validation Summary

Integral average values computed across a complete steady-state verification cycle:

Performance Metric Summary	MBGRN Solver	FEM Reference	Relative Error (%)
Average Mechanical Output Power	3120.76 W	3111.54 W	0.30%

5. Quantitative Performance Waveforms

5.1 Airgap Flux Density Spatial Distribution (Open-Circuit No-Load Component)

Radial Component (Br) Verification Metrics

Waveform Property	MBGRN Solver	FEM Reference	Discrepancy Error
Peak / Maximum Value	0.00 T	0.01 T	50.03%
Root-Mean-Square (RMS)	0.00 T	0.00 T	28.24%
Integral Mean / DC Component	0.00 T	-0.00 T	-100.00%
Peak-to-Peak Amplitude	0.00 T	0.01 T	Analytical Scale

Tangential Component (Bt) Verification Metrics

Waveform Property	MBGRN Solver	FEM Reference	Discrepancy Error
Peak / Maximum Value	0.25 T	0.23 T	8.12%
Root-Mean-Square (RMS)	0.11 T	0.11 T	5.26%
Integral Mean / DC Component	0.00 T	-0.00 T	-100.00%
Peak-to-Peak Amplitude	0.24 T	0.23 T	Analytical Scale

Axial Component (Bz) Verification Metrics

Waveform Property	MBGRN Solver	FEM Reference	Discrepancy Error
Peak / Maximum Value	0.74 T	0.73 T	1.09%
Root-Mean-Square (RMS)	0.62 T	0.61 T	0.58%
Integral Mean / DC Component	-0.00 T	0.00 T	100.00%
Peak-to-Peak Amplitude	0.74 T	0.73 T	Analytical Scale

Radial Component (Br) Spatial Harmonics Verification

Harmonic Order	MBGRN Amp (T)	FEM Amp (T)	Relative Error
Harmonic Order #1	0.0007	0.0005	49.72%

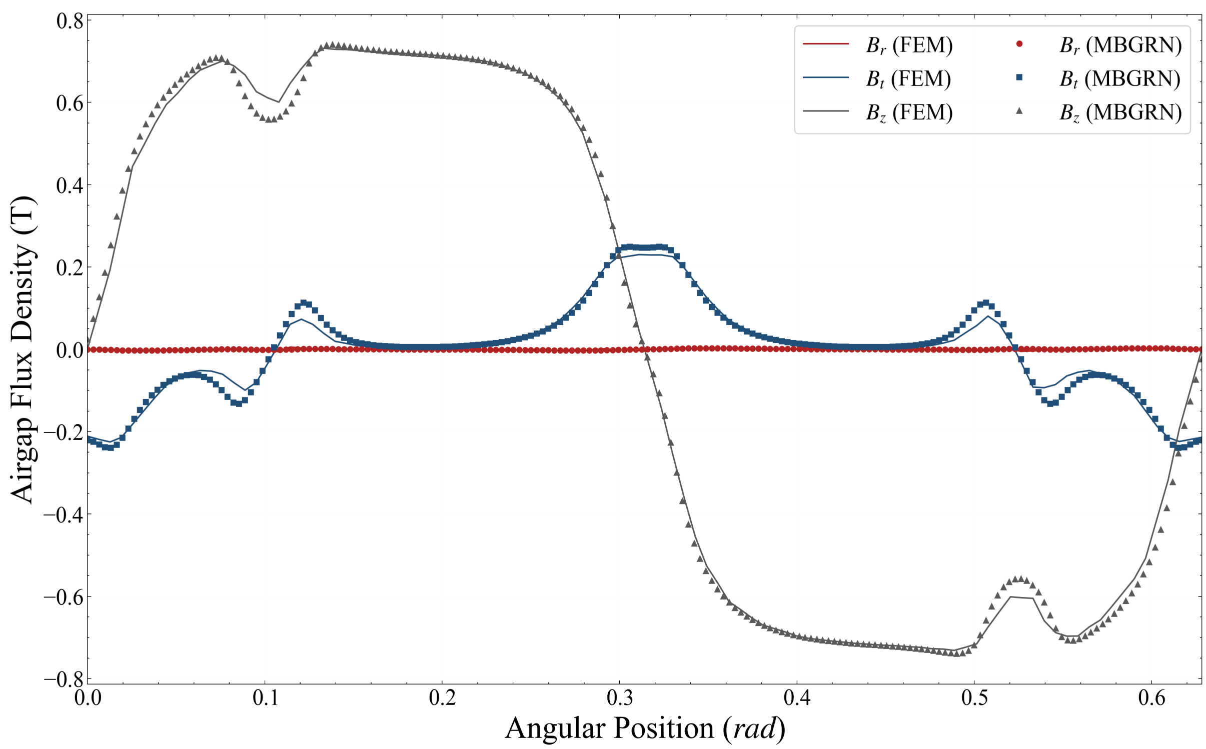
Harmonic Order #3	0.0015	0.0016	1.48%
Harmonic Order #5	0.0007	0.0010	26.32%
Harmonic Order #7	0.0005	0.0001	650.12%

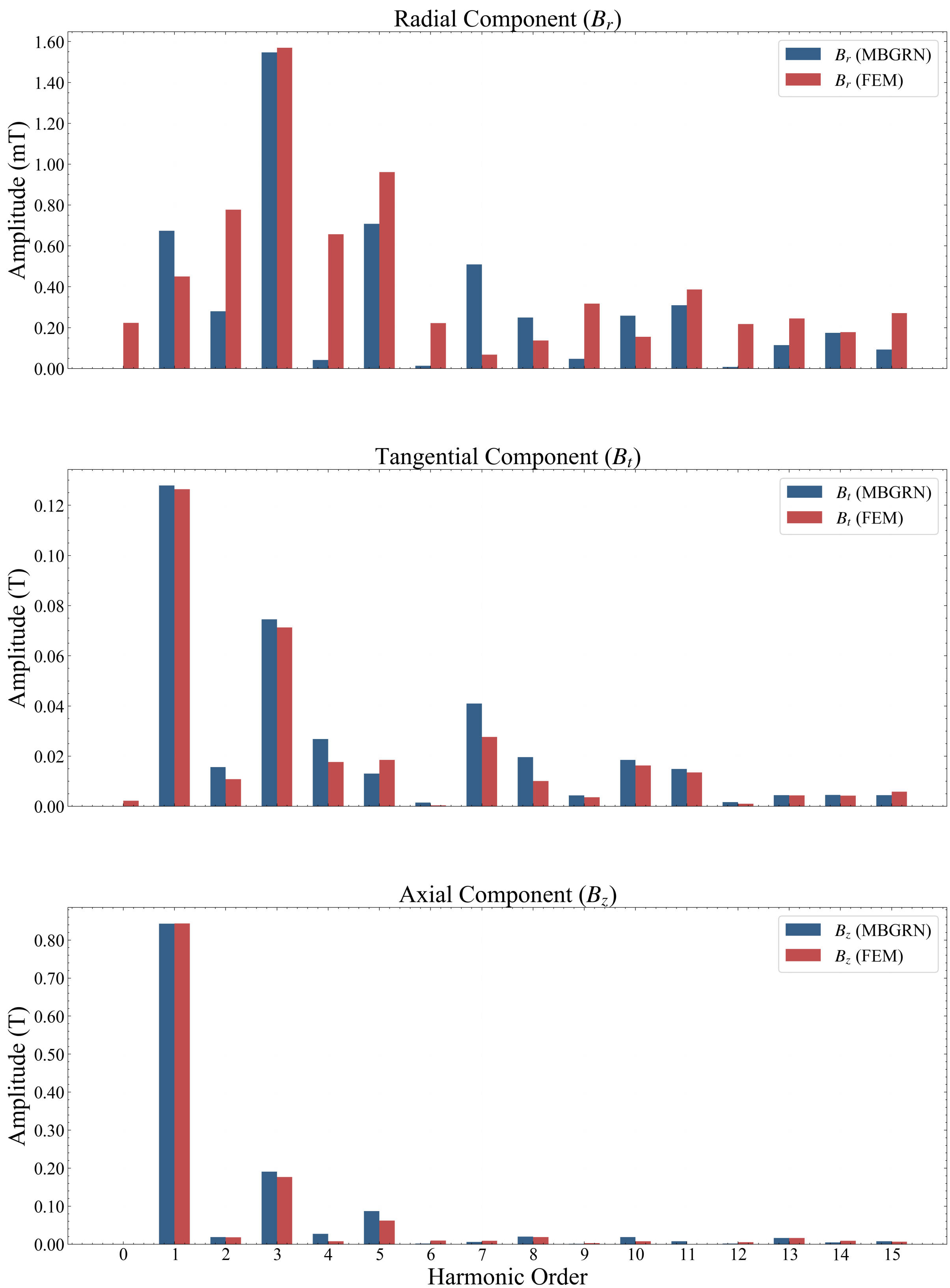
Tangential Component (Bt) Spatial Harmonics Verification

Harmonic Order	**MBGRN Amp (T)**	**FEM Amp (T)**	**Relative Error**
Harmonic Order #1	0.1279	0.1264	1.16%
Harmonic Order #3	0.0746	0.0713	4.59%
Harmonic Order #5	0.0130	0.0186	29.82%
Harmonic Order #7	0.0409	0.0277	47.97%

Axial Component (Bz) Spatial Harmonics Verification

Harmonic Order	**MBGRN Amp (T)**	**FEM Amp (T)**	**Relative Error**
Harmonic Order #1	0.8432	0.8437	0.07%
Harmonic Order #3	0.1906	0.1768	7.78%
Harmonic Order #5	0.0871	0.0621	40.20%
Harmonic Order #7	0.0062	0.0088	30.09%





Radial Component (B_r)
Amplitude (mT)
1.60
1.40
1.20
1.00
0.80
0.60
0.40
0.20
0.00
B_r (MBGRN)
B_r (FEM)
Tangential Component (B_t)
Amplitude (T)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
B_t (MBGRN)
B_t (FEM)
Axial Component (B_z)
Amplitude (T)
0.80
0.70
0.60
0.50
0.40
0.30
0.20
0.10
0.00
B_z (MBGRN)
B_z (FEM)
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Harmonic Order

5.2 Airgap Flux Density Spatial Distribution (On-Load Component)

Radial Component (Br) Verification Metrics

Waveform Property	MBGRN Solver	FEM Reference	Discrepancy Error
Peak / Maximum Value	0.00 T	0.01 T	64.34%
Root-Mean-Square (RMS)	0.00 T	0.00 T	32.43%
Integral Mean / DC Component	0.00 T	-0.00 T	-104.12%
Peak-to-Peak Amplitude	0.00 T	0.01 T	Analytical Scale

Tangential Component (Bt) Verification Metrics

Waveform Property	MBGRN Solver	FEM Reference	Discrepancy Error
Peak / Maximum Value	0.25 T	0.23 T	8.83%
Root-Mean-Square (RMS)	0.12 T	0.12 T	4.08%
Integral Mean / DC Component	0.00 T	-0.00 T	-100.00%
Peak-to-Peak Amplitude	0.25 T	0.23 T	Analytical Scale

Axial Component (Bz) Verification Metrics

Waveform Property	MBGRN Solver	FEM Reference	Discrepancy Error
Peak / Maximum Value	0.81 T	0.81 T	0.03%
Root-Mean-Square (RMS)	0.62 T	0.62 T	0.50%
Integral Mean / DC Component	0.00 T	0.00 T	70.24%
Peak-to-Peak Amplitude	0.82 T	0.81 T	Analytical Scale

Radial Component (Br) Spatial Harmonics Verification

Harmonic Order	MBGRN Amp (T)	FEM Amp (T)	Relative Error
Harmonic Order #1	0.0008	0.0009	14.01%
Harmonic Order #3	0.0015	0.0015	0.13%
Harmonic Order #5	0.0008	0.0012	28.09%
Harmonic Order #7	0.0006	0.0005	15.63%

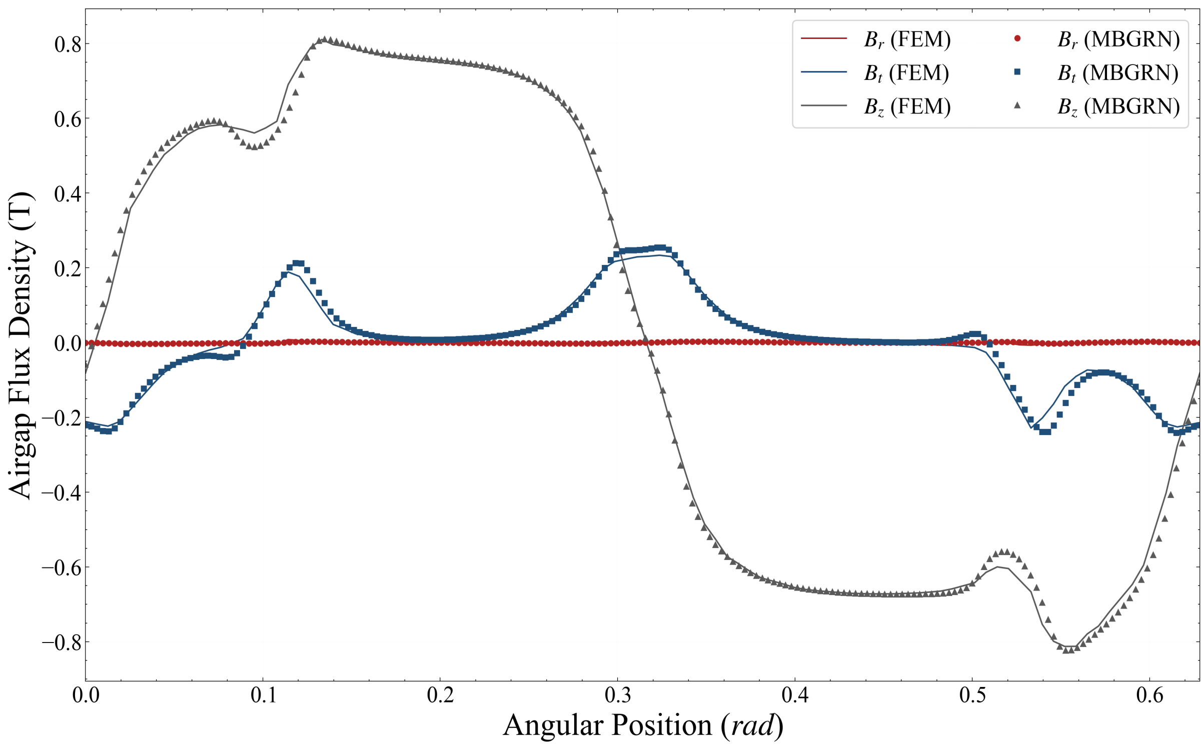
Tangential Component (Bt) Spatial Harmonics Verification

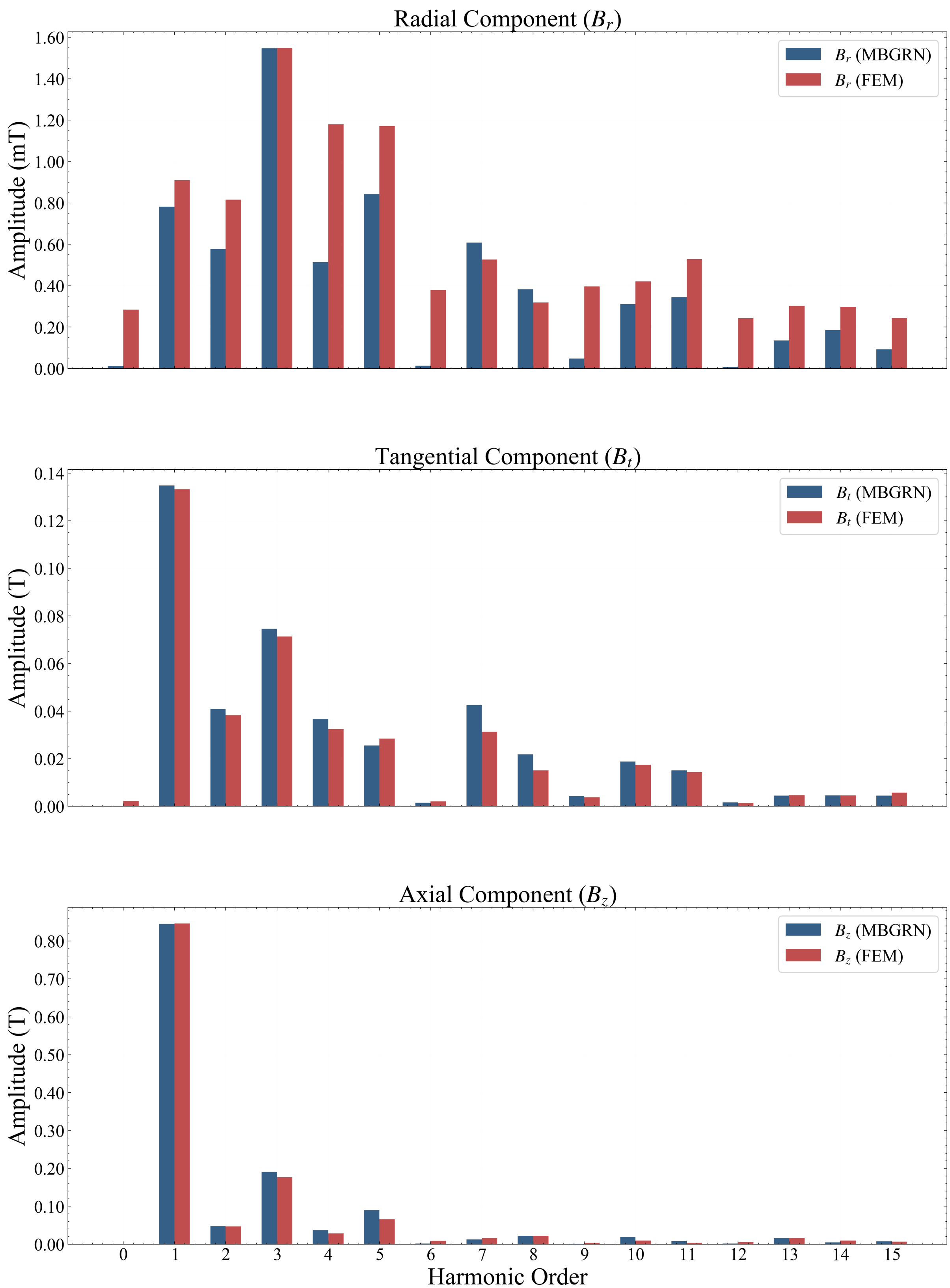
Harmonic Order	MBGRN Amp (T)	FEM Amp (T)	Relative Error
Harmonic Order #1	0.1348	0.1333	1.14%

Harmonic Order #3	0.0746	0.0714	4.55%
Harmonic Order #5	0.0256	0.0285	10.34%
Harmonic Order #7	0.0425	0.0313	35.57%

Axial Component (Bz) Spatial Harmonics Verification

Harmonic Order	**MBGRN Amp (T)**	**FEM Amp (T)**	**Relative Error**
Harmonic Order #1	0.8454	0.8465	0.13%
Harmonic Order #3	0.1905	0.1767	7.83%
Harmonic Order #5	0.0896	0.0662	35.44%
Harmonic Order #7	0.0125	0.0162	22.85%




Radial Component (B_r)
Amplitude (mT)
B_r (MBGRN)
B_r (FEM)
1.60
1.40
1.20
1.00
0.80
0.60
0.40
0.20
0.00
Tangential Component (B_t)
Amplitude (T)
B_t (MBGRN)
B_t (FEM)
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
Axial Component (B_z)
Amplitude (T)
B_z (MBGRN)
B_z (FEM)
0.80
0.70
0.60
0.50
0.40
0.30
0.20
0.10
0.00
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Harmonic Order

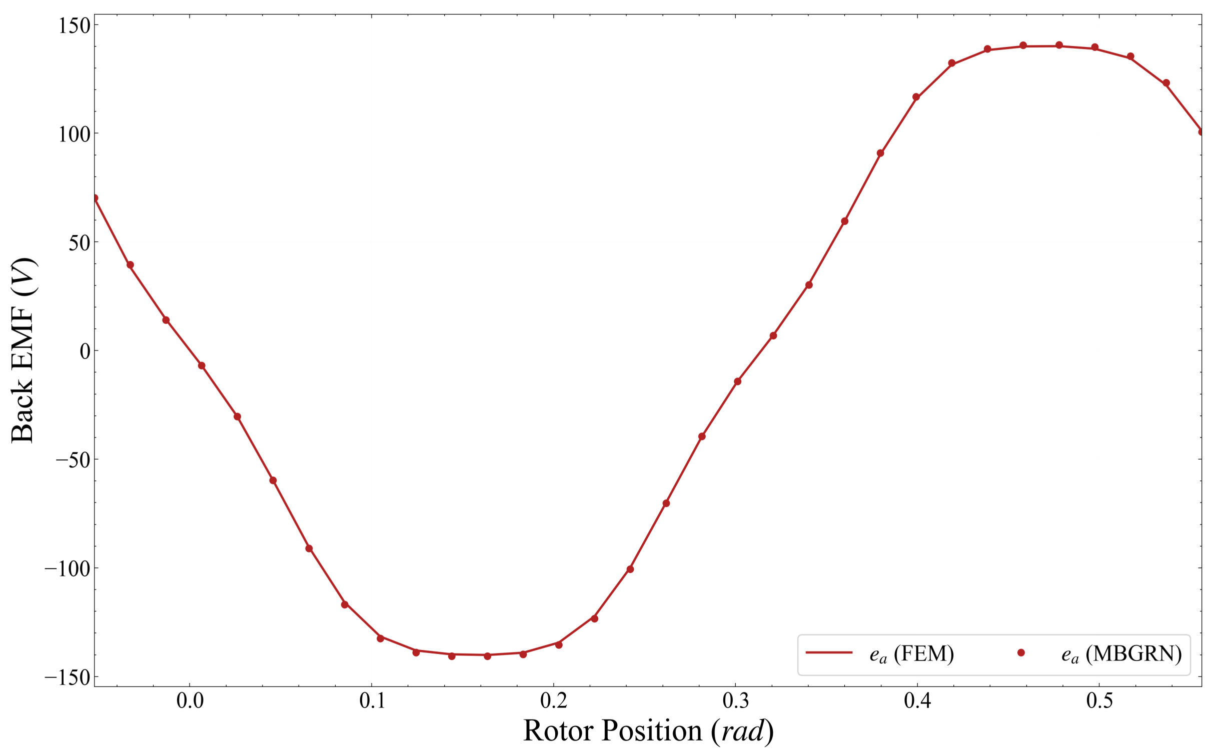
5.3 Phase Winding Back Electromotive Force Transient Waveforms (Open-Circuit No-Load Component)

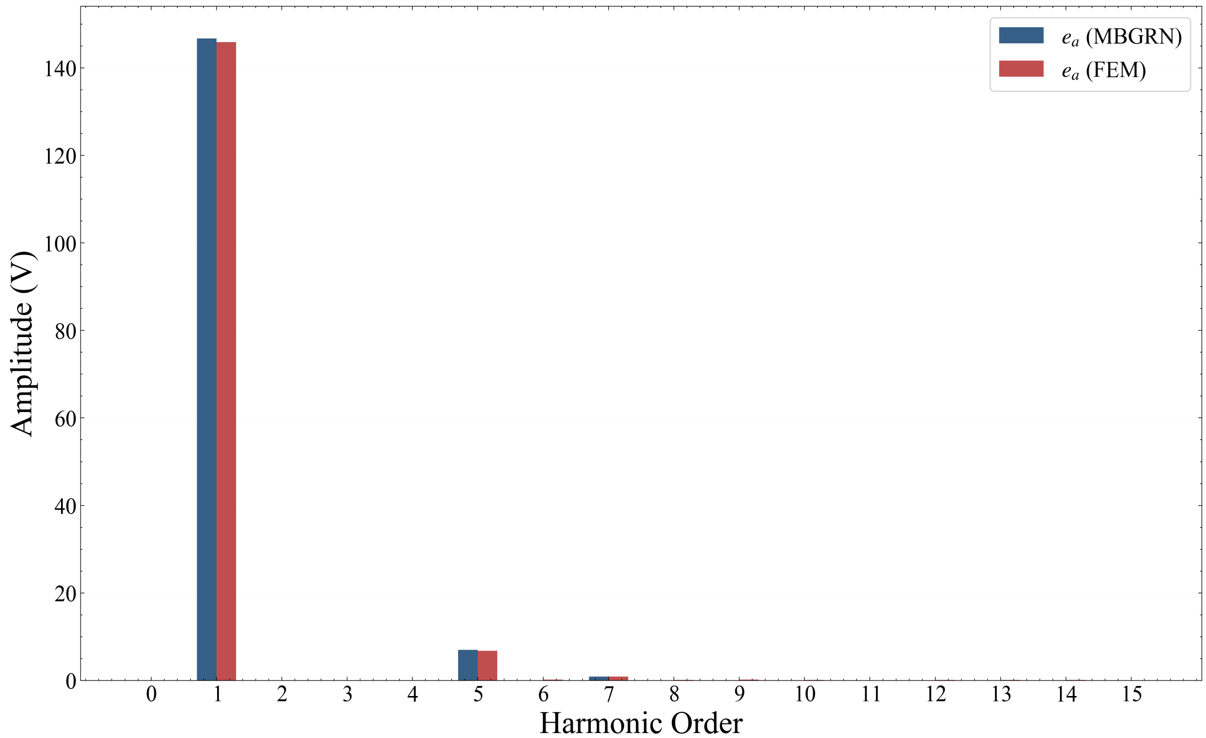
Phase-A Back EMF Verification Metrics

Waveform Property	MBGRN Solver	FEM Reference	Discrepancy Error
Peak / Maximum Value	140.63 V	140.06 V	0.41%
Root-Mean-Square (RMS)	103.89 V	103.30 V	0.56%
Integral Mean / DC Component	-0.00 V	-0.00 V	-66.67%
Peak-to-Peak Amplitude	140.63 V	140.08 V	Analytical Scale

Phase-A Back EMF Harmonics Verification

Harmonic Order	MBGRN Amp (V)	FEM Amp (V)	Relative Error
Harmonic Order #1	146.7467	145.9344	0.56%
Harmonic Order #3	0.0866	0.0783	10.54%
Harmonic Order #5	7.0218	6.7704	3.71%
Harmonic Order #7	0.8535	0.9129	6.51%





e_a (MBGRN)
e_a (FEM)
Amplitude (V)
0
20
40
60
80
100
120
140
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Harmonic Order

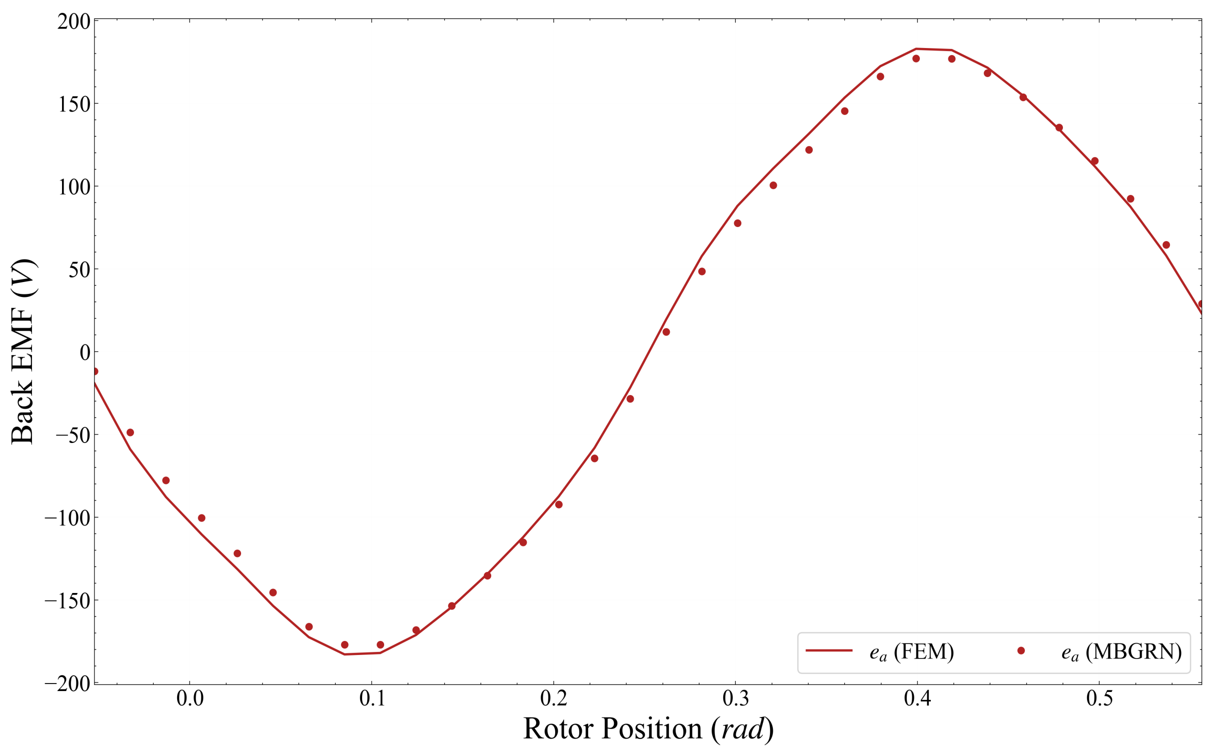
5.4 Phase Winding Back Electromotive Force Transient Waveforms (On-Load Component)

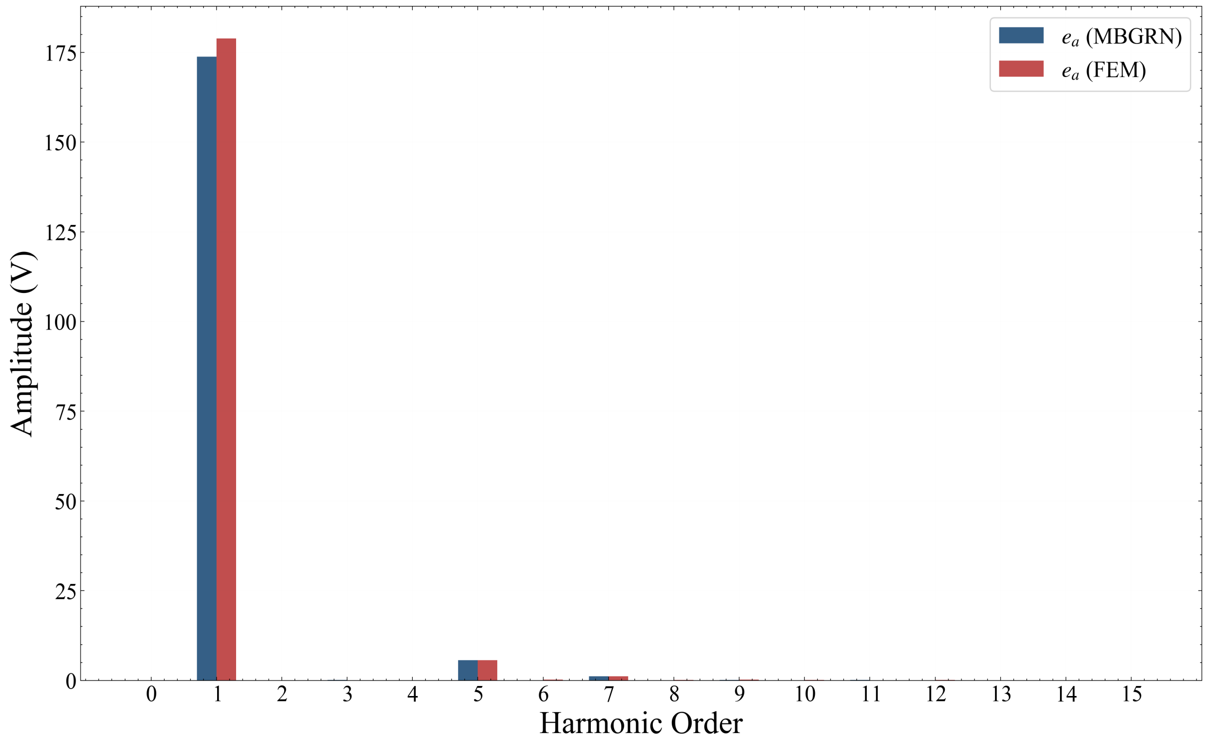
Phase-A Back EMF Verification Metrics

Waveform Property	MBGRN Solver	FEM Reference	Discrepancy Error
Peak / Maximum Value	177.12 V	182.86 V	3.14%
Root-Mean-Square (RMS)	122.97 V	126.55 V	2.83%
Integral Mean / DC Component	-0.00 V	0.00 V	1050.00%
Peak-to-Peak Amplitude	177.12 V	182.90 V	Analytical Scale

Phase-A Back EMF Harmonics Verification

Harmonic Order	MBGRN Amp (V)	FEM Amp (V)	Relative Error
Harmonic Order #1	173.8103	178.8741	2.83%
Harmonic Order #3	0.1216	0.0849	43.21%
Harmonic Order #5	5.6399	5.6585	0.33%
Harmonic Order #7	1.1836	1.1474	3.16%





e_a (MBGRN)
e_a (FEM)
Amplitude (V)
Harmonic Order
0
25
50
75
100
125
150
175
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15

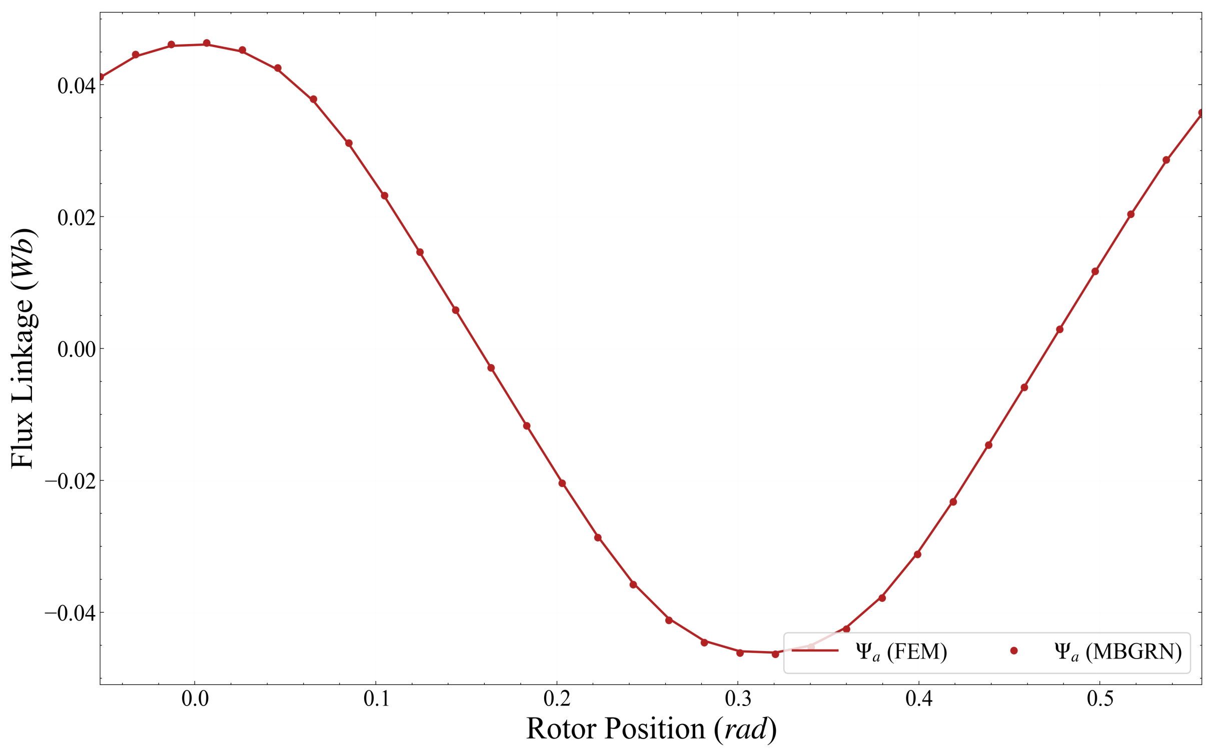
5.5 Core Winding Flux Linkage Transient Waveforms (Open-Circuit No-Load Component)

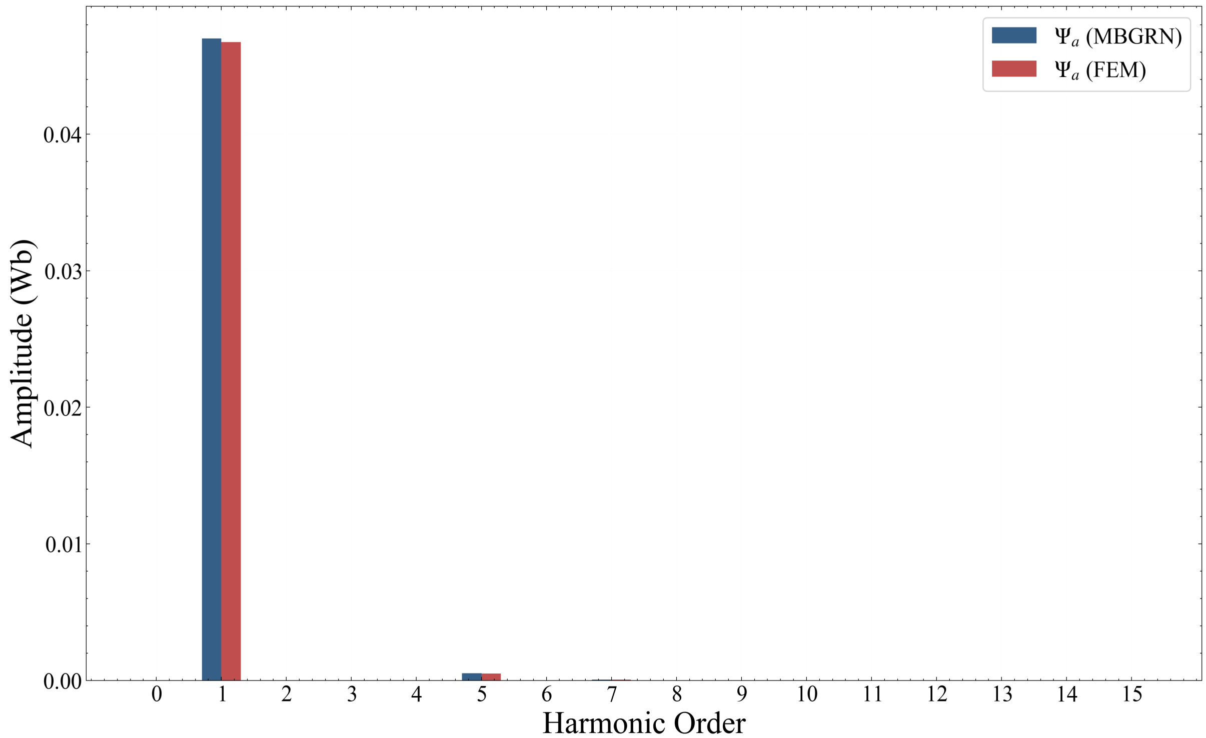
Phase-A Winding Flux Linkage Verification Metrics

Waveform Property	MBGRN Solver	FEM Reference	Discrepancy Error
Peak / Maximum Value	0.05 Wb	0.05 Wb	0.53%
Root-Mean-Square (RMS)	0.03 Wb	0.03 Wb	0.56%
Integral Mean / DC Component	-0.00 Wb	0.00 Wb	111.81%
Peak-to-Peak Amplitude	0.05 Wb	0.05 Wb	Analytical Scale

Phase-A Flux Linkage Spatial Harmonics Verification

Harmonic Order	MBGRN Amp (Wb)	FEM Amp (Wb)	Relative Error
Harmonic Order #1	0.0470	0.0468	0.56%
Harmonic Order #3	0.0000	0.0000	10.54%
Harmonic Order #5	0.0005	0.0005	3.71%
Harmonic Order #7	0.0001	0.0001	6.51%





Ψa (MBGRN)
Ψa (FEM)
Amplitude (Wb)
0.04
0.03
0.02
0.01
0.00
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Harmonic Order

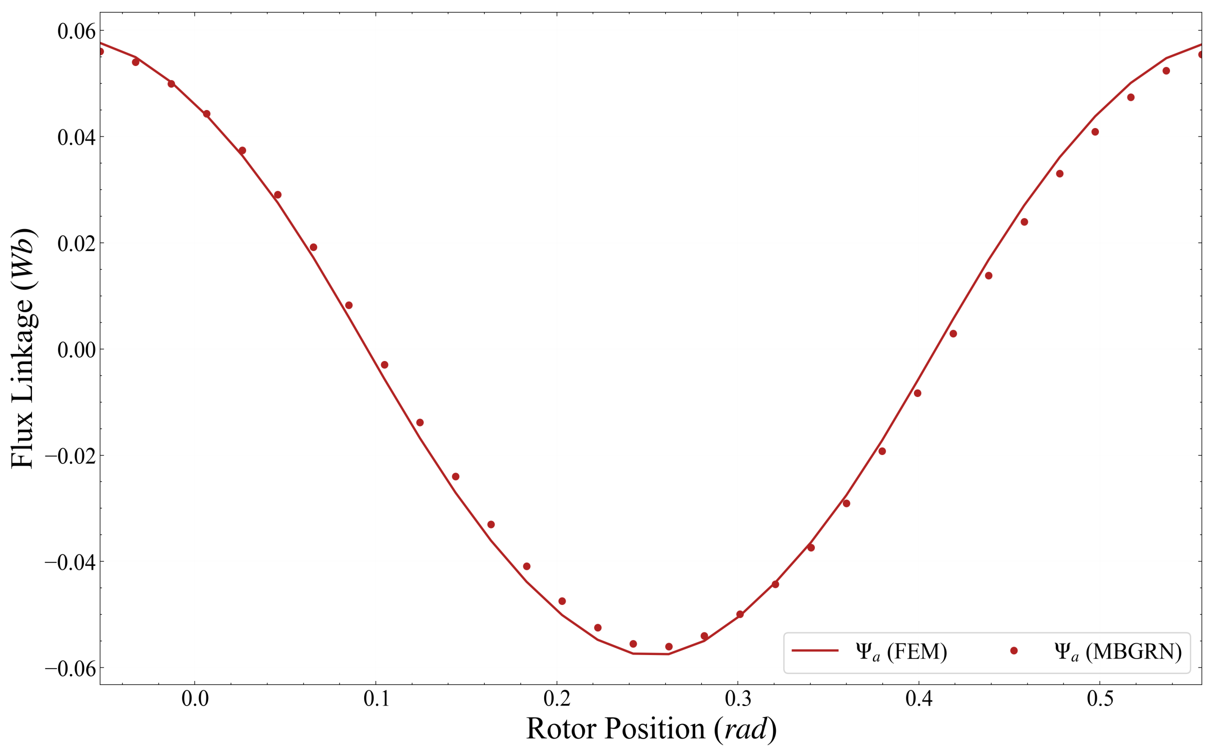
5.6 Core Winding Flux Linkage Transient Waveforms (On-Load Component)

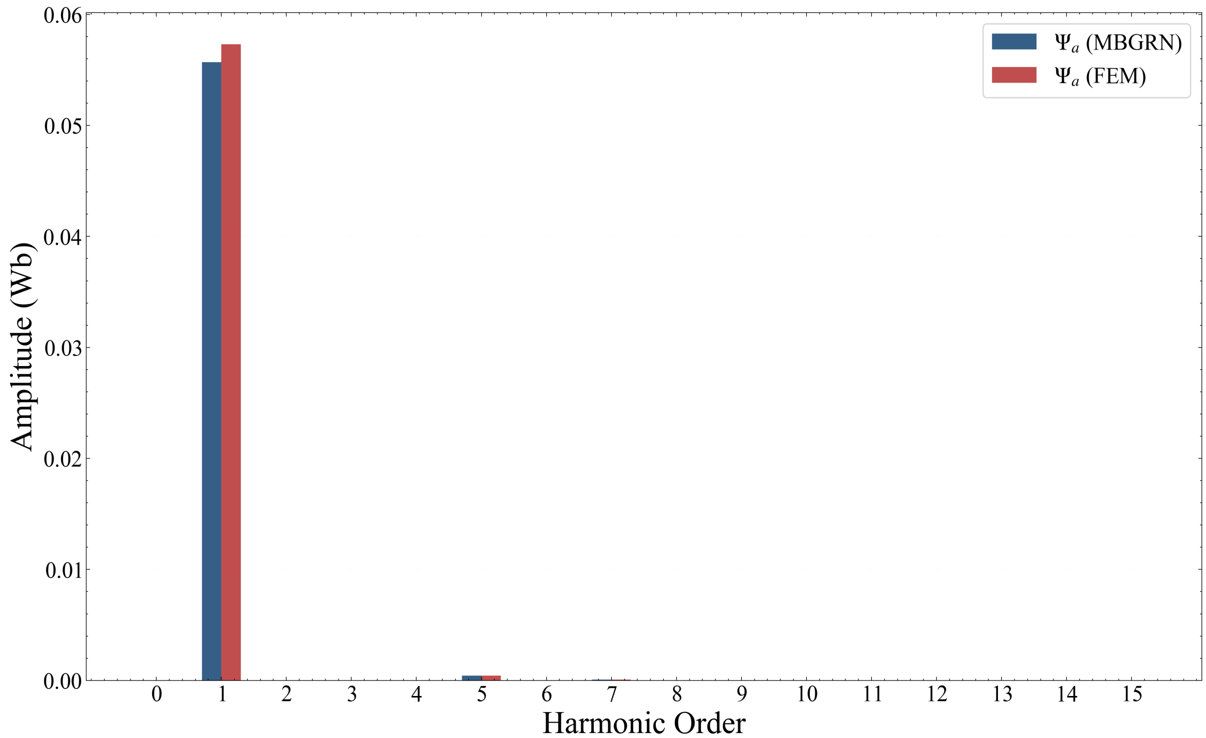
Phase-A Winding Flux Linkage Verification Metrics

Waveform Property	MBGRN Solver	FEM Reference	Discrepancy Error
Peak / Maximum Value	0.06 Wb	0.06 Wb	2.74%
Root-Mean-Square (RMS)	0.04 Wb	0.04 Wb	2.83%
Integral Mean / DC Component	0.00 Wb	0.00 Wb	10.71%
Peak-to-Peak Amplitude	0.06 Wb	0.06 Wb	Analytical Scale

Phase-A Flux Linkage Spatial Harmonics Verification

Harmonic Order	MBGRN Amp (Wb)	FEM Amp (Wb)	Relative Error
Harmonic Order #1	0.0557	0.0573	2.83%
Harmonic Order #3	0.0000	0.0000	43.21%
Harmonic Order #5	0.0004	0.0004	0.33%
Harmonic Order #7	0.0001	0.0001	3.16%





Ψ_a (MBGRN)
Ψ_a (FEM)
Amplitude (Wb)
0.06
0.05
0.04
0.03
0.02
0.01
0.00
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Harmonic Order

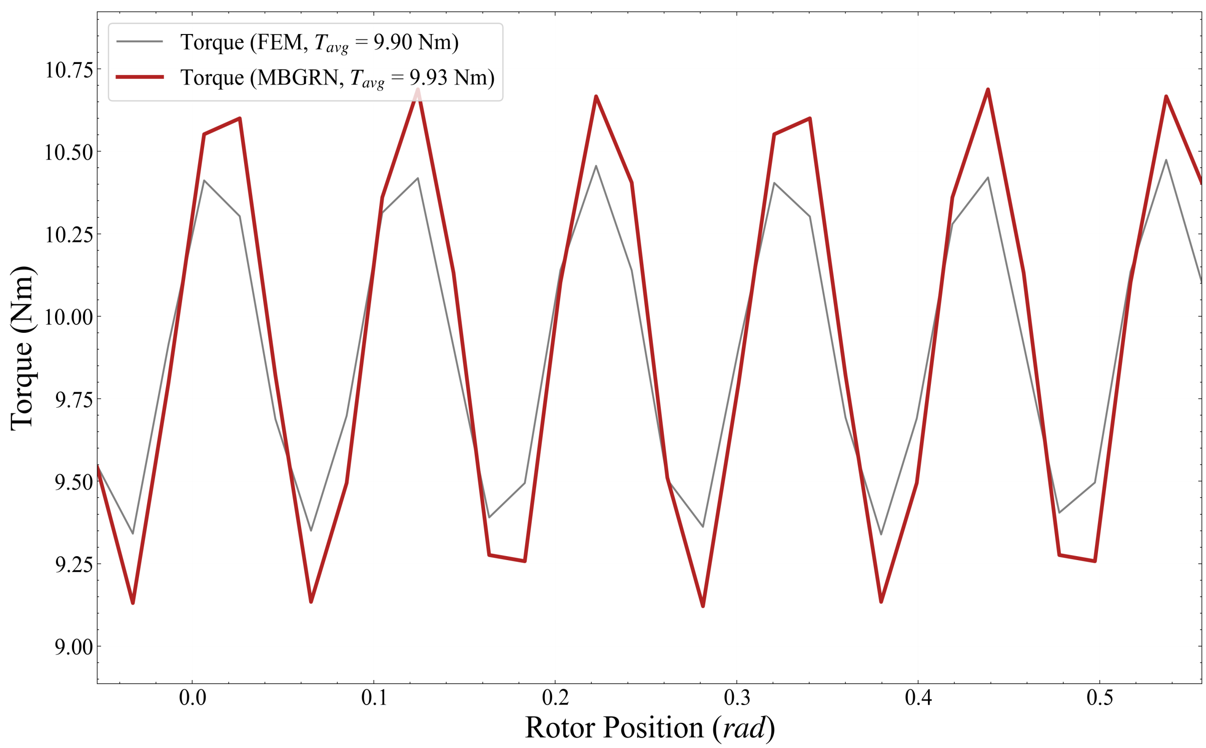
5.1 Electromagnetic Torque Transient Performance Profiles

Electromagnetic Torque Quantitative Verification Metrics

Waveform Property	MBGRN Solver	FEM Reference	Discrepancy Error
Peak / Maximum Value	10.69 Nm	10.47 Nm	2.04%
Root-Mean-Square (RMS)	9.95 Nm	9.91 Nm	0.37%
Integral Mean / DC Component	9.93 Nm	9.90 Nm	0.30%
Peak-to-Peak Amplitude	0.78 Nm	0.57 Nm	Analytical Scale
Torque Ripple Ratio (%)	15.77%	11.46%	37.57%

Electromagnetic Torque Spectrum Harmonics Verification

Harmonic Order	MBGRN Amp (Nm)	FEM Amp (Nm)	Relative Error
Harmonic Order #1	0.0026	0.0030	12.27%
Harmonic Order #3	0.0025	0.0026	5.49%
Harmonic Order #5	0.0023	0.0043	46.01%
Harmonic Order #7	0.0021	0.0025	19.36%



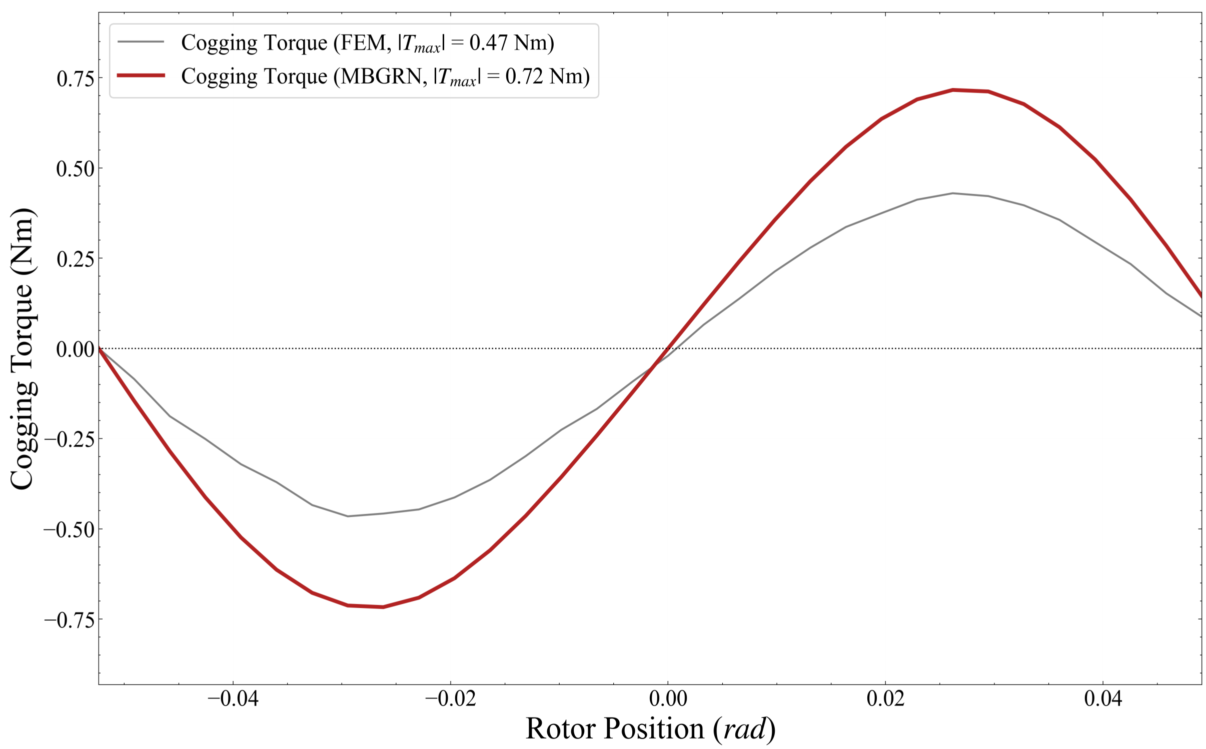
5.2 Open-Circuit No-Load Cogging Torque Waveforms Profile

Cogging Torque Quantitative Verification Metrics

Waveform Property	MBGRN Solver	FEM Reference	Discrepancy Error
Peak / Maximum Value	0.72 Nm	0.43 Nm	66.61%
Root-Mean-Square (RMS)	0.50 Nm	0.31 Nm	62.72%
Integral Mean / DC Component	-0.00 Nm	-0.01 Nm	97.43%
Peak-to-Peak Amplitude	0.72 Nm	0.45 Nm	Analytical Scale

Cogging Torque Spectrum Harmonics Verification

Harmonic Order	MBGRN Amp (Nm)	FEM Amp (Nm)	Relative Error
Harmonic Order #1	0.7074	0.4345	62.83%
Harmonic Order #3	0.0088	0.0097	9.59%
Harmonic Order #5	0.0004	0.0040	90.28%
Harmonic Order #7	0.0000	0.0033	99.31%



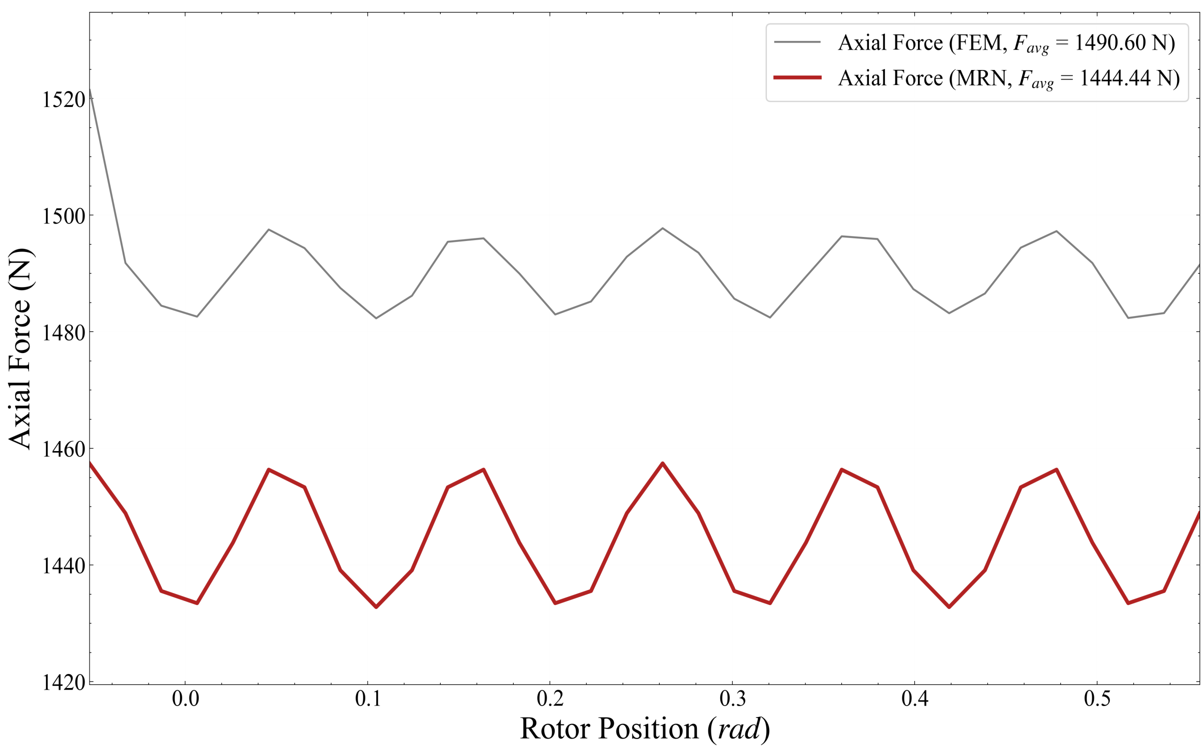
5.7 Axial Force Transient Waveforms (Open-Circuit No-Load Component)

Axial Force Quantitative Verification Metrics

Waveform Property	MBGRN Solver	FEM Reference	Discrepancy Error
Peak / Maximum Value	1457.43 N	1521.47 N	4.21%
Root-Mean-Square (RMS)	1444.47 N	1490.62 N	3.10%
Integral Mean / DC Component	1444.44 N	1490.60 N	3.10%
Peak-to-Peak Amplitude	12.33 N	19.59 N	Analytical Scale

Axial Force Spectrum Harmonics Verification

Harmonic Order	MBGRN Amp (N)	FEM Amp (N)	Relative Error
Harmonic Order #1	0.0040	1.2227	99.68%
Harmonic Order #3	0.0040	1.1288	99.65%
Harmonic Order #5	0.0040	1.2827	99.69%
Harmonic Order #7	0.0040	1.9011	99.79%



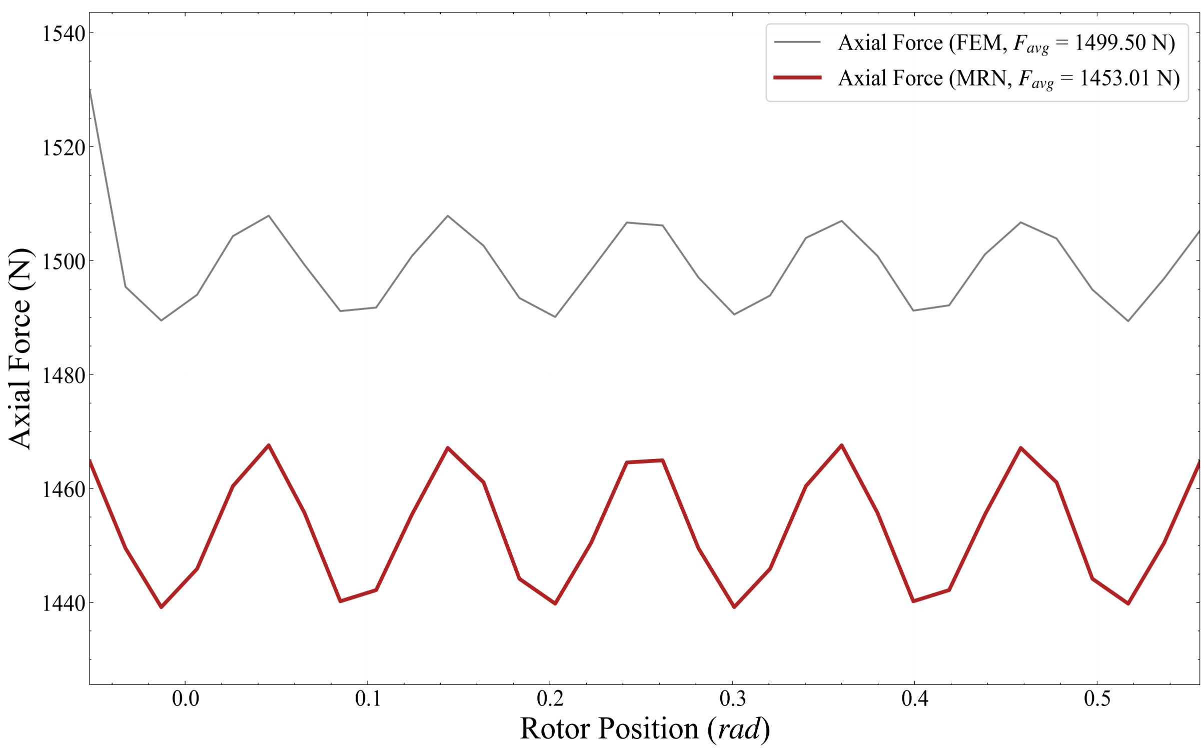
5.8 Axial Force Transient Waveforms (On-Load Component)

Axial Force Quantitative Verification Metrics

Waveform Property	MBGRN Solver	FEM Reference	Discrepancy Error
Peak / Maximum Value	1467.59 N	1529.97 N	4.08%
Root-Mean-Square (RMS)	1453.05 N	1499.52 N	3.10%
Integral Mean / DC Component	1453.01 N	1499.50 N	3.10%
Peak-to-Peak Amplitude	14.20 N	20.29 N	Analytical Scale

Axial Force Spectrum Harmonics Verification

Harmonic Order	MBGRN Amp (N)	FEM Amp (N)	Relative Error
Harmonic Order #1	0.0117	1.2151	99.04%
Harmonic Order #3	0.0115	1.2111	99.05%
Harmonic Order #5	0.0114	1.2799	99.11%
Harmonic Order #7	0.0113	1.8610	99.39%



5. Computational Performance & Resource Utilization

Quantitative comparison of solver execution speed, numerical scale, and computational resource efficiency:

Performance Metric Summary	**MBGRN Solver**	**FEM Reference**	**Comparison Ratio**
Total Computation Execution Time	1132.420 s	14984.7 s	13.2x faster
Memory used	1432.2 MB	3143.7 MB	2.2x lower
Total Elements	61,248	164,964	2.7x fewer
System matrix size	61,247	223,695	3.7x smaller